DOA estimation based on combined ESPRIT for co-prime array

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***Abstract*— The issue of direction of arrival (DOA) estimation for co-prime array is studied, and an method based on combined estimation of signal parameters via rotational invariance technique (ESPRIT) is proposed. ESPRIT is employed to obtain an ambiguous estimation, which then can be used to recover all the other estimations. Then the DOA estimation can be uniquely determined by finding the coincide estimations from the two subarrays. Compared to combined multiple signal classification (MUSIC) method, the proposed algorithm can achieve better estimation performance with much lower complexity. Simulation results verify the effectiveness of the proposed algorithm.**

Keywords*— *co-prime array; DOA estimation; ESPRIT; combined

I. INTRODUCTION

Direction of arrival (DOA) estimation is an important issue for array signal processing. Well-known algorithms such as multiple signal classification (MUSIC) method [1]-[2] and estimation of signal parameters via rotational invariance technique (ESPRIT) [3]-[4] have been proposed for DOA estimation. However, the compact arrays they used limit the estimation performance and suffer from mutual coupling problem [5]. Recently, co-prime array has attracted lots of attention for it can achieve large degrees of freedom (DOF) with large inter-element spacing [6]. DOA estimation based on combined MUSIC [7]-[8] has been proved superior to conventional methods, but it need a peak search involves high complexity.

In this paper, a DOA estimation algorithm using co-prime array based on combined ESPRIT is proposed. Due to the large inter-element spacing, ambiguous estimations are obtained via ESPRIT. Then DOA can be uniquely estimated by finding the coincide results from the two subarrays. Compared to combined MUSIC, the proposed algorithm can achieve better DOA estimation performance but much lower complexity.

II. DATA MODEL

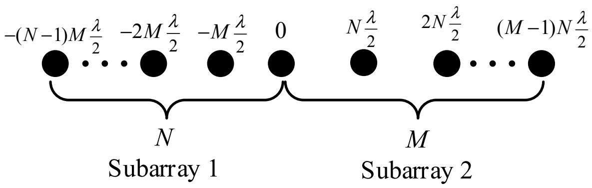

Fig. 1. Proposed co-prime array structure for DOA estimation

By sharing the same reference element, two subarrays of the co-prime array are aligned along the positive axis and negative axis, respectively. The element coordinates are shown in Fig.1, where M and N are co-prime integers and λ denotes the wavelength. Assume that there are K far-field sources, then the output of the array can be expressed as

$$\mathbf{x}(t) = [\mathbf{a}(\theta_1), \mathbf{a}(\theta_2)\cdots, \mathbf{a}(\theta_K)]\mathbf{s}(t) + \mathbf{n}(t) \quad (1)$$

where θ_k is the DOA of the kth target with respect to the array normal. $\mathbf{s}(t) = [s_1(t), s_2(t), \cdots, s_K(t)]^T$ is the source vector. $\mathbf{n}(t)$ is an additive white Gaussian noise (AWGN) vector. $\mathbf{a}(\theta_k)$ is the steering vector for the k-th target, which is $\mathbf{a}(\theta_k) = [e^{-j(N-1)M\pi\sin\theta_k}, \cdots, 1, e^{jN\pi\sin\theta_k}, \cdots, e^{j(M-1)N\pi\sin\theta_k}]^T$.

III. DOA ESTIMATION BASED ON COMBINED ESPRIT

A. Ambiguous DOA estimation based on ESPRIT

The covariance matrix is estimated via

$$\mathbf{R} = \frac{1}{J}\sum_{t=1}^{J}\mathbf{x}(t)\mathbf{x}^H(t) \quad (2)$$

where J is the snapshot number. The K eigenvectors of $\mathbf{R}$ corresponding to the K largest eigenvalues form the signal subspace $\mathbf{E}_s$ [3], whose first N rows and last M rows are denoted as $\mathbf{E}_{sN}$ and $\mathbf{E}_{sM}$, respectively. $\mathbf{E}_{sN}$ is corresponding to subarray 1, and it satisfies

$$\mathbf{E}_{SN1}\mathbf{T}^{-1}\mathbf{\Phi}_M\mathbf{T} = \mathbf{E}_{SN2} \quad (3)$$

where $\mathbf{E}_{SN1}$ and $\mathbf{E}_{SN2}$ are the first and last (N-1) rows of $\mathbf{E}_{SN}$, respectively. $\mathbf{\Phi}_M = diag(e^{jM\pi\sin\theta_1}, e^{jM\pi\sin\theta_2}, \cdots, e^{jM\pi\sin\theta_K})$, and $\mathbf{T}$ is a $K \times K$ nonsingular matrix [3]. Thus the eigenvalues of $\mathbf{E}_{SN1}{}^{+}\mathbf{E}_{SN2}$ ($\lambda_{Mk}, k = 1, ..., K$) are the diagonal elements of $\mathbf{\Phi}_M$, and the DOA information can be extracted,

$$\sin\breve{\theta}_{k,m} = angle(\lambda_{Mk})/(M\pi), k = 1, ..., K \quad (4)$$

where $angle(\cdot)$ is the phase extraction function, whose result is within the range [-π, π]. Therefore, $\sin\breve{\theta}_{k,m}$ is not a correct estimation but an ambiguous estimation within the range [-1/M, 1/M], which will be used to recover all estimations within [-1, 1] in the next section. Similarly, use $\mathbf{E}_{SM1}$ and $\mathbf{E}_{SM2}$ to denote the first and last (M-1) rows of $\mathbf{E}_{SM}$, respectively. Then,

$$\mathbf{E}_{SM1}\mathbf{T}^{-1}\mathbf{\Phi}_N\mathbf{T} = \mathbf{E}_{SM2} \tag{5}$$

where $\mathbf{\Phi}_N = diag(e^{jN\pi\sin\theta_1}, e^{jN\pi\sin\theta_2}, \cdots, e^{jN\pi\sin\theta_K})$. As $\mathbf{T}^{-1}$ has been obtained from the eigenvectors of $\mathbf{E}_{SN1}{}^{+}\mathbf{E}_{SN2}$ previously, then $\mathbf{\Phi}_N$ can be estimated via $\mathbf{\Phi}_N = \mathbf{T}\mathbf{E}_{SM1}{}^{+}\mathbf{E}_{SM2}\mathbf{T}^{-1}$, whose diagonal elements $\lambda_{Nk}, k = 1,...,K$ give the ambiguous DOA information

$$\sin\widehat{\theta}_{k,n} = angle(\lambda_{Nk})/(N\pi), k = 1,...,K \tag{6}$$

Due to the same column permutation $\mathbf{T}^{-1}$, estimations obtained from Eq.(4) and Eq.(6) are corresponding to the same DOA.

B. *Unique DOA determination*

Firstly, we need to recover all estimation results based on the ambiguous estimations obtained via Eq.(4) and Eq.(6). According to [7]-[8], the adjacent intervals between the estimations are 2/M and 2/N for subarray 1 and subarray 2, respectively. Base on this relationship, all the M estimations $\sin\breve{\theta}_{k,m}, m = 1,...,M$ for subarray 1 and N estimations $\sin\widehat{\theta}_{k,n}$, $n = 1,...,N$ for subarray 2 can be obtained.

Then the correct estimation can be obtained by finding the coincide one or the average of two nearest ones from $\sin\breve{\theta}_{k,m}, m = 1,...,M$ and $\sin\widehat{\theta}_{k,n}, n = 1,...,N$ [6]-[8]. Finally, the unique DOA estimation can be obtained.

The proposed algorithm can achieve closed-form solutions of the DOA estimation without peak search, so it has much lower complexity than Combined MUSIC [7]-[8].

IV. SIMULATION RESULTS

Define root mean square error (RMSE) as

$$RMSE = \frac{1}{K}\sum_{k=1}^{K}\sqrt{\left(\frac{1}{L}\sum_{l=1}^{L}[(\hat{\theta}_{k,l} - \theta_k)^2]\right)} \tag{7}$$

where $\hat{\theta}_{k,l}$ is the estimate of DOA θ_k of the lth Monte Carlo trial. Assume that there are K = 2 independent sources with the DOAs being $(\theta_1, \theta_2) = (20^\circ, 35^\circ)$. M=6, N=5 and J=128 snapshots are collected.

Fig.2 shows the DOA estimation performance comparison between unitary ESPRIT [4], unitary Root MUSIC [2], Combined MUSIC [7] and the proposed one. Benefit from the usage of co-prime array, Combined MUSIC and the proposed algorithm both have much better estimation performance than the other two. The proposed algorithm has slightly better estimation performance than Combined MUSIC, which has much higher complexity.

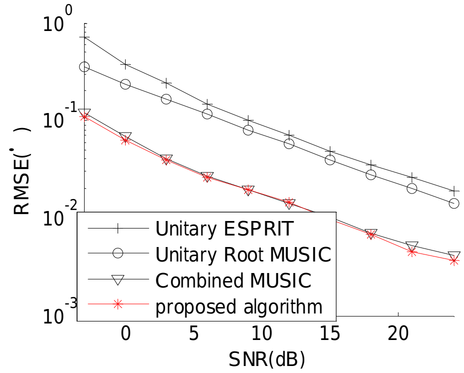

Fig. 2. DOA estimation performance comparison versus SNR

V. CONCLUSION

A DOA estimation algorithm using co-prime array based on combined ESPRIT is proposed. By recovering all estimations based on the ambiguous estimations obtained via ESPRIT, the unique DOA estimation can be achieved by finding the coincide estimations from the two subarrays. The proposed algorithm can achieve better DOA estimation performance but much lower complexity than combined MUSIC.

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